

BÀI TOÁN PHÂN LỚP

Các giả định sau được sử dụng trong GLM:

- Mỗi quan hệ giữa các biến phụ thuộc và độc lập có thể là phi tuyến tính;
- Biến phụ thuộc có thể có phân phối không chuẩn;
- Để ước tính các tham số chưa biết, ta sử dụng phương pháp ước lượng hợp lý cực đại (maximum likelihood estimation- MLE), không sử dụng OLS.
- Các sai số là độc lập nhưng có thể có phân phối không chuẩn;

$$f(y_i) = \exp\left(\frac{y_i\theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi)\right) = e^{\frac{y_i\theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi)}$$

$$E(Y_i) = \mu_i = b'(\theta_i)$$

$$Var(Y_i) = \sigma^2_i = b''(\theta_i) a_i(\phi)$$

The Normal distribution

$$f(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{(y_i - \mu_i)^2}{\sigma^2}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{y_i^2 + \mu_i^2 - 2y_i\mu_i}{\sigma^2}\right) = \exp\left(\frac{y_i\mu_i - \frac{1}{2}\mu_i^2}{\sigma^2} - \frac{y_i^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)\right)$$

- $\theta_i = \mu_i$;
- $\phi = \sigma^2$;
- $a_i(\phi) = \phi$;
- $b_i(\theta_i) = \frac{1}{2}\theta_i^2$;
- $c(y_i, \phi) = -\frac{y_i^2}{2\phi} - \frac{1}{2}\log(2\pi\phi)$

The Binomial distribution: Phân bố nhị thức

$$f_i(y_i) = \mathbb{P}(Y_i = y_i) = \binom{N_i}{y_i} p_i^{y_i} (1 - p_i)^{N_i - y_i}, \quad y_i = 0, 1, \dots, N_i$$

$$\log (f_i (y_i)) = y_i \log (p_i) + (N_i - y_i) \log (1 - p_i) + \log \left(\binom{N_i}{y_i} \right)$$

$$= y_i \log \left(\frac{p_i}{1 - p_i} \right) + N_i \log (1 - p_i) + \log \left(\binom{N_i}{y_i} \right)$$

- $\theta_i = \log \left(\frac{p_i}{1 - p_i} \right);$

- $p_i = \frac{\exp(\theta_i)}{1 + \exp(\theta_i)}$ and $1 - p_i = \frac{1}{1 + \exp(\theta_i)}$

- $b(\theta_i) = N_i \log(1 + \exp(\theta_i))$

- $c(y_i, \phi) = \log \left(\binom{N_i}{y_i} \right)$

- $a_i(\phi) = \phi$ and $\phi = 1.$

Distribution of Y	Link function name	Link function	Model	Prediction, $\hat{\mu}$	Domain of Y
Normal	Identity	$g(\mu) = \mu$	$\mu = \mathbf{X}\beta$	$\hat{\mu} = \mathbf{X}\hat{\beta}$	$Y \in (-\infty, +\infty)$
Exponential	Negative Inverse	$g(\mu) = -\mu^{-1}$	$-\mu^{-1} = \mathbf{X}\beta$	$\hat{\mu} = -(\mathbf{X}\hat{\beta})^{-1}$	$Y \in (0, +\infty)$
Poisson	Log	$g(\mu) = \log(\mu)$	$\log(\mu) = \mathbf{X}\beta$	$\hat{\mu} = \exp(\mathbf{X}\hat{\beta})$	$Y \in \{0, 1, 2, \dots\}$
Binomial	Logit	$g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$	$\log\left(\frac{\mu}{1-\mu}\right) = \mathbf{X}\beta$	$\hat{\mu} = \frac{\exp(\mathbf{X}\hat{\beta})}{1 + \exp(\mathbf{X}\hat{\beta})}$	$Y \in \{0, 1\}$

Hồi quy logistics- Phân bố Bernoulli

$$\mathbb{P}(Y_i = y_i) = p_i^{y_i} (1 - p_i)^{1-y_i}, \quad y_i \in \{0, 1\}$$

$$\begin{aligned}\mathbb{E}(Y_i) &= \mu_i = p_i \\ \text{Var}(Y_i) &= \sigma_i^2 = p_i (1 - p_i)\end{aligned}$$

$$\log \left(\frac{p_i}{1 - p_i} \right) = \mathbf{X}_i \boldsymbol{\beta}$$

Maximum Likelihood Estimation (Ước lượng hợp lý cực đại)

- Lập hàm hợp lý:

$$\mathcal{L}(\boldsymbol{\beta} | \mathbf{Y} = \mathbf{y}, \mathbf{X} = \mathbf{x}) = \prod_{i=1}^N \mathbb{P}(Y_i = y_i | \mathbf{X} = \mathbf{x}, \boldsymbol{\beta})$$

$$\mathbb{P}(Y_1 = y_1, \dots, Y_N = y_N) = \prod_{i=1}^N p_i^{y_i} (1 - p_i)^{1-y_i}$$

$$\begin{aligned}
\ell(\boldsymbol{\beta}|\mathbf{Y} = \mathbf{y}, \mathbf{X} = \mathbf{x}) &= \log(\mathcal{L}(\boldsymbol{\beta}|\mathbf{Y} = \mathbf{y}, \mathbf{X} = \mathbf{x})) = \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \\
&= \sum_{i=1}^N \left[y_i \log\left(\frac{p_i}{1 - p_i}\right) + \log(1 - p_i) \right] = \sum_{i=1}^N [y_i \cdot \mathbf{x}_i \boldsymbol{\beta} - \log(1 + \exp(\mathbf{x}_i \boldsymbol{\beta}))]
\end{aligned}$$

$$\frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^N \left[y_i x_j - \frac{1}{1 + \exp(\mathbf{x}_i \boldsymbol{\beta})} \exp(\mathbf{x}_i \boldsymbol{\beta}) x_j \right] = \sum_{i=1}^N \left[\left(y_i - \frac{\exp(\mathbf{x}_i \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i \boldsymbol{\beta})} \right) x_j \right] = 0$$

$$\beta^{(n+1)} = \beta^{(n)} - \frac{f'(\beta^{(n)})}{f''(\beta^{(n)})}$$

$$\log\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right)=\widetilde{\mathbf{X}}_i\hat{\boldsymbol{\beta}}$$

$$\hat{p}_i\left(\widetilde{\mathbf{X}}\right)=\frac{\exp\left(\widetilde{\mathbf{X}}_i\widehat{\boldsymbol{\beta}}\right)}{1+\exp\left(\widetilde{\mathbf{X}}_i\widehat{\boldsymbol{\beta}}\right)}$$

$$\hat{p}_i(\mathbf{X}) \in \left[\frac{\exp\left(\mathbf{X}\hat{\boldsymbol{\beta}} - z_c \times se(\mathbf{X}\hat{\boldsymbol{\beta}})\right)}{1 + \exp\left(\mathbf{X}\hat{\boldsymbol{\beta}} - z_c \times se(\mathbf{X}\hat{\boldsymbol{\beta}})\right)}; \frac{\exp\left(\mathbf{X}\hat{\boldsymbol{\beta}} + z_c \times se(\mathbf{X}\hat{\boldsymbol{\beta}})\right)}{1 + \exp\left(\mathbf{X}\hat{\boldsymbol{\beta}} + z_c \times se(\mathbf{X}\hat{\boldsymbol{\beta}})\right)} \right]$$

$$se(\mathbf{X}\hat{\boldsymbol{\beta}}) = \sqrt{\text{diag}\left(\text{Var}(\mathbf{X}\hat{\boldsymbol{\beta}})\right)}.$$

$$\mathbb{V}\mathrm{ar}\left(\mathbf{X}\hat{\boldsymbol{\beta}}\right)=\mathbf{X}\mathbb{V}\mathrm{ar}\left(\hat{\boldsymbol{\beta}}\right)\mathbf{X}^{\top}$$

Hồi quy Probit

$$p_i = \mathbb{P}(Y_i = 1 | \mathbf{X}_i) = \mathbb{P}(Y_i \leq \mathbf{X}_i \beta) = \Phi(\mathbf{X}_i \beta)$$

$$\Phi^{-1}\left(p_i\right)=X_i\beta$$

Confusion Matrix (Contingency Table)

		Predicted	
		1	0
Actual	1	True Positive	False Negative
	0	False Positive	True Negative

$$\textit{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{True Negative} + \text{False Positive} + \text{False Negative}}$$

$$\text{Misclassification Error} = \frac{\text{False Positive} + \text{False Negative}}{\text{True Positive} + \text{True Negative} + \text{False Positive} + \text{False Negative}}$$

- The **True Positive Rate (TPR)** - It indicates how many positive values, out of all the positive values, have been **correctly predicted**:

$$\text{TPR} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

This is also known as **Sensitivity** or **Recall**.

- The **True Negative Rate (TNR)** - It indicates how many negative values, out of all the negative values, have been **correctly predicted**:

$$\text{TNR} = \frac{\text{True Negative}}{\text{True Negative} + \text{False Positive}}$$

This is also known as **Specificity**.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

$$F_1 = 2. \frac{\text{Precision.TPR}}{\text{Precision+TPR}}$$

$$F_\beta = \left(1 + \beta^2\right). \frac{\text{Precision.TPR}}{\beta^2.\text{Precision+TPR}}$$

